Kitchcu

CPO Product Blueprint

Modules | Functionalities | Data Flows | Application Flows | Pain Points | Solutions | Roadmap

Kitchcu cloud kitchen platform

Version 3.0 | July 2026 | Confidential

Product North Star

Kitchcu is the operating system for cloud kitchens - not an aggregator. Subscription SaaS with zero food commission. Owners run everything from one PWA; customers trust what they order through live media and home-taste ratings.

WhatsApp to revenue	**Live photos + fair deliv**	**Subscription, 0% food**	**10 pilot kitchens / 3 mo**
Owner promise	Customer promise	Platform model	MVP target

Non-Negotiable Principles

- Truth in media - hero dish photos must be live-capture (no stock deception)
- Quality over speed - owner-set prep/delivery windows, never fake 10-min races
- Owner sovereignty - CRM, coupons, and customer history belong to the kitchen
- Progressive complexity - advanced features unlock after kitchen traction (50+ orders)
- WhatsApp-native - meet owners where 70%+ of orders already happen

Owner Pain Points (Market Reality)

Pain	Business Impact	Severity
Orders trapped in WhatsApp chats	Lost orders, zero analytics	Critical
Aggregator 18-30% commission	Margin erosion, no customer ownership	Critical
No daily profit visibility	Cash flow guesswork, bad decisions	High
Stock photos mislead customers	Refunds, reputation damage	High
Inconsistent taste batch-to-batch	Repeat customer loss	High
No CRM or targeted marketing	Cannot win back regulars	Medium
Promotions based on guesswork	Wasted discounts	Medium
Walk-in + call + chat silos	Double booking, chaos	Medium

Customer Pain Points

Pain	Why It Matters	Kitchcu Answer
Cannot trust menu photos	Expectation vs reality gap	Live-capture dish media
Opaque delivery fees	Cart abandonment	PostGIS distance + owner rules
Generic star ratings	No home-food signal	Home-taste benchmark 1-5
One kitchen per cart	Inconvenient lunch combos	Multi-kitchen checkout (P2)
No order transparency	Anxiety, support calls	Lifecycle + WhatsApp updates
No tiffin subscription	Daily meal friction	Kitchen meal plans (P2)

"I run 80 orders a day on WhatsApp and still do not know my profit." - Typical owner

Pain Point to Solution Mapping

Pain ID	Problem	Kitchcu Module	Solution
P1	WhatsApp chaos	Order + Notification	Unified inbox, parser, manual fallback
P2	Aggregator commission	Billing	Flat subscription; kitchen keeps food revenue
P3	No profit view	Analytics	Revenue, dish, pattern reports
P4	False menu photos	Catalog + Media	Live-capture enforced for hero images
P5	Taste inconsistency	Catalog + Ingredient	Per-dish standards, stock mapper (P3)
P6	No customer ownership	Marketing	Owner CRM, coupons, tiffin
P7	Guesswork promos	Growth Engine	Seasonal, win-back, combo suggestions
P8	Multi-channel chaos	Order Service	Single lifecycle for all sources

Platform Architecture (Microservices)

```
      +------------------+
                   |   API Gateway    |  :18000  (path-based routing)
                   +--------+---------+
                            |
       +--------------------+--------------------+
       v                    v                    v
+-------------+     +-------------+     +-------------+
|  Identity   |     |   Catalog   |     |    Order    |   ... Billing, Notification
|  S1 DONE    |     |  S2 DONE    |     |  S3 DONE    |
+------+------+     +------+------+     +------+------+
       |                   |                   |
       +-------------------+-------------------+
                           v
             +---------------------------+
             |  PostgreSQL 16 + PostGIS  |
             |  schema-per-domain (ACID) |
             +-------------+-------------+
                           |
             +-------------+-------------+
             v                           v
       Redis Streams                 MinIO/S3
       (event bus + cache)           (dish media)
```

Event-driven design: every write publishes EventEnvelope after DB commit.

Platform Modules & Build Status

Module	Responsibility	Schema	Status
Gateway	Auth routing, service proxy	-	DONE v0.3
Identity	Owners, kitchens, OTP, JWT	ckac_identity	DONE S1
Catalog	Dishes, categories, live media	ckac_catalog	DONE S2
Order	Manual orders, lifecycle, history	ckac_orders	DONE S3
Notification	WhatsApp, push, SMS, support chat & tickets	ckac_support	S4 LIVE
Billing	Subscriptions, UPI, split pay	ckac_billing	Sprint 6
Analytics	Owner revenue, segments, peak hours	order svc	S5 PARTIAL
Marketing	CRM, coupons, tiffin	ckac_marketing	Phase 2
Rating	Home taste, A/V reviews	ckac_ratings	Phase 2
Delivery	Radius, fees, tracking	geo rules	Phase 2
Growth	AI-ready suggestions	ckac_growth	Phase 2-3
Learning / Stream	Recipes, live prep	ckac_learning	Phase 3

Module: Identity Service

Functionalities

- Owner registration (phone + OTP)
- JWT access + refresh tokens
- Kitchen onboarding with PostGIS location
- Kitchen code generation (e.g. CKPNQ001)
- Owner-kitchen authorization checks

API Endpoints (via Gateway)

- POST /api/v1/owners/register
- POST /api/v1/auth/otp/request | verify

- POST /api/v1/kitchens
- GET /api/v1/kitchens/me

Data Model (ckac_identity)

- owners: id, phone, name, subscription_tier, status
- kitchens: id, owner_id, code, name, location (geography)

Module: Catalog Service

Functionalities (F13-F15)

- 10 default categories seeded per kitchen
- Dish CRUD with price, prep time, quality notes
- Live-capture rule: hero image must be is_live_capture
- Public menu endpoint (cacheable)
- Publishes dish.created / dish.updated events

Business Rules

- No stock photo as hero - CPO trust mandate
- Cross-schema read: verify kitchen ownership
- No cross-schema writes
- Menu invalidation on dish.updated (Redis)

Events

dish.created -> ckac:catalog:dish stream

Module: Order Service

Functionalities (F03-F05, F30)

- Manual order creation (walk-in, phone)
- Line items from catalog (price snapshot)
- Lifecycle state machine with audit trail
- Order history with status/source filters
- Per-dish prep time drives ETA

Order Code Format

- {kitchen_code}-BILL-{YYYYMMDD}-{SEQ}
- Example: CKPNQ001-BILL-20260712-0001
- Bill ID: BILL-20260712-0001

Lifecycle States

```
received -> accepted -> preparing -> ready -> out_for_delivery -> delivered
Any pre-delivered state -> cancelled (reason required)
Each transition -> order_status_events row + order.status.changed event
```

Planned Modules (Sprints 4-6 & Phase 2)

Module	Key Features	Events / Integrations
Notification S4	WhatsApp webhook, message parser, push	Consumes order.*; F01-F02, F45
Billing S6	Owner subscription, UPI, COD, Razorpay	payment.captured; F26, F42-F44
Analytics S6+	Revenue, best dishes, patterns	Consumes order.delivered; F07-F12
Marketing P2	CRM, coupons, tiffin, daily push	F34-F40
Delivery P2	Radius, fee quote, tracking links	PostGIS; F27-F31
Rating P2	Home taste + A/V anonymous reviews	F16-F18
Growth P2	Seasonal, win-back, combo suggestions	F11

Owner Application Flow - Day 1 to Daily Ops

```
      DAY 1 ONBOARDING
  Register phone -> OTP (JWT) -> Create kitchen (geo pin)
  -> Add 5 dishes (live camera) -> Publish menu
  -> Connect WhatsApp Business (S4) -> First order in inbox

DAILY OPERATIONS
  Home Inbox: new orders + WhatsApp drafts + today stats
  -> Tap order -> Confirm / edit items -> Advance lifecycle (1 tap)
  -> Customer notified (WhatsApp + push)
  -> End of day: revenue report, top dish, growth tip

OWNER PWA NAV (target)
  Home | Orders (Active/History/Drafts) | Menu | Customers | Analytics
  | Marketing | Subscriptions | Payments | Settings
```

Customer Application Flow (Phase 2 PWA)

```
     DISCOVERY & ORDER
  Open PWA link or Discover map (PostGIS nearby)
  -> Filter: distance, rating, live-prep premium
  -> Kitchen profile: live photos, home-taste scores
  -> Add items (multi-kitchen cart supported)
  -> Delivery fee breakdown shown -> Pay UPI or COD
  -> Track: received ... delivered (WhatsApp + push)

POST-ORDER
  -> Rate home taste (1-5) + quality + optional 15s video
  -> Repeat order in 2 taps from history
  -> Subscribe to tiffin plan (optional)

CUSTOMER PWA NAV
  Discover | Cart | Orders & Tracking | Subscriptions | Ratings | Profile
```

Order Intake - Multi-Source Data Flow

```
       SOURCES                    INTAKE PATH                         OUTPUT
---------                  -----------                         ------
WhatsApp message    ->    Webhook -> Parser -> Draft order  ->  Owner confirms
Phone / walk-in     ->    Owner manual entry (S3 DONE)      ->  order.placed
Customer PWA        ->    Cart checkout (Phase 2)           ->  order.placed

order.placed event:
  DB commit (ckac_orders) -> Redis Stream ckac:orders:order
  -> Notification service (owner alert + customer confirm)
  -> Analytics worker (counters, daily stats)
  -> Billing worker (payment capture when online)
```

Order Lifecycle - Event-Driven Data Flow

```
       Owner taps status advance (PATCH /orders/{id}/status)
        |
        v
Validate transition (state machine) -> Update orders.status
        |
        v
Insert order_status_events (from, to, note, created_by)
        |
        v
Publish order.status.changed -> ckac:orders:order
        |
   +----+----+----+
   v    v    v    v
 Push  WA   CRM  Analytics
 (F45)      update  (future)
```

Cancel requires reason. Terminal states: delivered, cancelled.

Event Catalog (Event-Driven Design)

Event	Producer	Stream	Consumers
kitchen.created	identity	ckac:identity:kitchen	analytics
dish.created	catalog	ckac:catalog:dish	search, cache
dish.updated	catalog	ckac:catalog:dish	menu cache invalidation
order.placed	order	ckac:orders:order	notify, analytics, billing
order.status.changed	order	ckac:orders:order	notify, tracking UI
payment.captured	billing	ckac:billing:payment	settlement, reports
whatsapp.message.received	notification	ckac:notify:wa	order parser

Contract: DB commit first, then publish. Outbox table in ckac_events for reliability.

Database Architecture - Schema-per-Domain

Schema	Tables	Phase	Status
ckac_identity	owners, kitchens	1	LIVE
ckac_catalog	categories, dishes, dish_media	1	LIVE
ckac_orders	orders, order_items, order_status_events	1	LIVE
ckac_events	outbox, processed_events	1	LIVE
ckac_billing	payments, subscriptions, settlements	1-2	Planned S6
ckac_marketing	kitchen_customers, coupons, plans	2	Planned
ckac_ratings	dish_ratings, suggestions	2	Planned
ckac_growth	suggestions, seasonal_patterns	2-3	Planned

Scaling: 0-500 kitchens single PG; read replica + PgBouncer at 500-5K; partition events by month at 5K+.

API Gateway Routing & Service Ports

Path Pattern	Target Service	Port (host)
/api/v1/auth/*, /api/v1/owners/*	Identity	18001
/api/v1/kitchens/{id}/categories|menu|dishes	Catalog	18002
/api/v1/kitchens/{id}/orders/*	Order	18003
/api/v1/orders/*	Order	18003
/api/v1/kitchens (create/list)	Identity	18001
Gateway entry	Gateway	18000

Phase 1 Features - Owner Can Run Kitchen

ID	Feature	Module	Sprint	Status
F01	WhatsApp order capture	notification	S4	PARTIAL
F02	Message parser	notification	S4	DONE
F03	Manual order input	order	S3	DONE
F04	Order lifecycle	order	S3	DONE
F05	Order history	order	S3	DONE
F07	Revenue report	analytics	S6	Planned
F13-F15	Menu + live photo + categories	catalog	S2	DONE
F26	Owner subscription	billing	S6	Planned
F30	Per-dish prep time	order+catalog	S3	DONE
F42-F43	UPI / COD payments	billing	S6	Planned
F45	WhatsApp + app notifications + AI support	notification	S4	PARTIAL

Phase 2 Features - Growth (Months 4-6)

Customer Experience

- F06 Multi-kitchen single checkout
- F32 Kitchen discovery (map + distance)
- F33 Order history + repeat order
- F16-F18 Home taste ratings + A/V
- F41 Custom cooking requests

Owner Growth

- F34-F35 Tiffin / meal plans
- F36-F38 CRM, coupons, targeted pricing

- F39 Daily menu WhatsApp push
- F07-F11 Analytics + growth suggestions

- F44 Aggregated payment + split settlement

- F27-F31 Delivery radius, fees, tracking

Phase 3 Features - Differentiation (Moat)

- F19 Ingredient balance mapper - stock deduct, low-stock alerts
- F20 Customer recipe suggestions (owner accept/reject workflow)
- F21-F23 Learning portal, trial batches, recipe rewards
- F24 Best cloud chef rankings (city / state / national)
- F46-F48 Live kitchen streaming (optional owner opt-in premium)
- F12 Performance report with recipe improvement suggestions

Ranking Formula (Benchmark)

Monthly score = 30% avg rating + 20% recipe shares + 25% review volume + 15% repeat rate + 10% consistency index.

Current Build Status - July 2026

S1-S4 complete; S5 pa
Sprints done

5 (+ gateway)
Services live

90+ passing
Automated tests

Portal + 3 PWAs
Apps live

Sprint	Deliverable	Status
S1	Identity, gateway, Docker, JWT, TDD foundation	COMPLETE
S2	Catalog, live-capture, dish.created events	COMPLETE
S3	Order manual, lifecycle, order.placed events	COMPLETE
S4	WhatsApp webhook + parser + AI support chat	COMPLETE
S5	PWAs + portal + analytics + ticketing	PARTIAL LIVE
S6	Billing + revenue reports	Planned

Multi-Kitchen Checkout Flow (F06 - Phase 2)

```
     Customer cart:
[Kitchen A: 2 dishes] + [Kitchen B: 1 dish]
            |
            v
     master_orders (1 customer payment)
            |
    +-------+-------+
    v               v
order (CKA-...)   order (CKB-...)
separate lifecycle, separate bills, separate tracking
            |
            v
Razorpay Route: automatic split to each kitchen linked account
Customer: unified receipt + per-kitchen tracking tabs
```

Delivery Fee Decision Flow (F27-F28)

```
1. Compute distance (PostGIS) owner kitchen <-> customer address
2. If distance <= free_delivery_radius -> fee = 0
3. If > radius -> apply owner rules (flat / per-km / min order)
4. Show breakdown at checkout BEFORE payment
5. Customer denies fee:
   a. Owner can cancel order
   b. Owner can waive (min order rule)
   c. Customer switches to self-pickup
```

Owner Growth Intelligence Engine (F11)

Suggestion Type	Trigger	Example
Seasonal	Calendar + regional patterns	Diwali in 3 weeks - add mithai combo
Dish promo	High rating, low volume	Paneer Tikka 4.8 but 5% orders - promote
Win-back	Regular inactive 21+ days	12 customers - send coupon MAAS20
Combo	Order association mining	Naan + Dal ordered together 68% - bundle
Peak staffing	Historical volume by hour	Fri 12-2 PM is 3x avg - prep extra

Rating & Trust Layer (F16-F18)

Dimension	Scale	Purpose
Home taste	1-5	Tastes like authentic home cooking
Quality	1-5	Freshness, portion, packaging
Optional A/V	15-30 sec	Anonymous video/audio review

- Ratings only from verified completed orders - no fake reviews
- Aggregated on dish page without customer identity
- Feeds chef rankings and growth suggestions

Business Model - Subscription, Not Commission

Tier	Price/mo	Includes
Starter	Rs 499	WhatsApp orders, menu, basic reports
Pro	Rs 1,499	CRM, marketing, tiffin, coupons, analytics
Enterprise	Rs 3,999	Live stream, API, multi-branch

- Zero per-order food commission - kitchen keeps 100% food revenue
- Customer tiffin: 0% Kitchcu cut
- Payments: pass-through via Razorpay Route (multi-kitchen split)
- Yearly plans: 2 months free

CPO Success Metrics & KPIs

Metric	Phase 1 Target	How Measured
Time to first order	< 5 min post-signup	Onboarding funnel
WhatsApp parse rate	95% parsed or manual	Parser accuracy log
Lifecycle notify latency	< 2s event to push	Event SLA monitoring
Hero live-capture rate	100%	Catalog audit
Owner retention M6	80%	Subscription churn
Repeat customer rate	40%+	CRM aggregates
Owner GMV on platform	North star	Sum of order totals

Competitive Position

- vs Aggregators (Swiggy/Zomato): zero commission, owner owns CRM, live-capture integrity
- vs POS (PetPooja): customer-facing discovery, quality trust layer, WhatsApp analytics
- Unique combination: WhatsApp-native ops + live menu + multi-kitchen cart + home-taste ratings
- Kitchcu occupies high owner-control quadrant WITH customer discovery - not either/or

Rs 2.5L Cr+

TAM India delivery GMV

~50,000

SAM semi-formal kitchens

500 kitchens

SOM Year 3 target

Rs 9 Cr

ARR at Rs 1,499/mo

Go-To-Market Strategy

- Pilot clusters: 10 kitchens in one locality - network effect for customer discovery
- WhatsApp-first onboarding in Hindi and regional languages
- Kitchen-branded PWA links shared on Instagram / WhatsApp status
- Monthly city chef rankings for organic PR and community buzz
- Aggregator escape narrative: Keep 100% of your food revenue
- Progressive feature unlock - simple Day 1, powerful Month 3

Technology Stack (Scalable Foundation)

Layer	Choice	Rationale
Frontend	PWA React + Vite + TS	WhatsApp deep links, one codebase
Backend	Python FastAPI microservices	Async, OpenAPI, event-driven
Database	PostgreSQL 16 + PostGIS	ACID, geo queries, schema isolation
Cache / Events	Redis Streams	Menu cache, event bus (Kafka Phase 4)
Media	MinIO / S3	Live-capture dish photos
Payments	Razorpay Route	Multi-kitchen split settlement
Notify	WhatsApp Cloud API + Web Push	Dual channel reach
Infra	Docker -> Kubernetes	500 kitchens today, 50K path

Every cloud kitchen deserves to operate like a brand

- Data instead of guesswork - revenue, patterns, growth suggestions
- Trust instead of stock photos - live capture + home-taste ratings
- Direct relationships instead of rented customers - owner CRM
- Community instead of isolation - chef rankings, recipe rewards

Kitchcu is infrastructure - not the next food aggregator.

Shipped Module Status (July 2026)

Module	Challenge solved	Status
Identity + Catalog + Order	Onboard, live menu, lifecycle	DONE S1-S3
Notification + Support	WhatsApp + tickets + tracking nudges	DONE S4/S14
Billing + GST finance	Subscription + monthly tax audit	DONE S6+GST
Multi-kitchen + Route pay	One cart, fair kitchen payouts	DONE S8-S9
Marketing CRM / coupons	Owner owns customers	DONE S10
Ratings + Growth	Home taste + actionable suggestions	DONE S11-S12
Delivery + Ingredients	Fair fees + pantry deduct	DONE S13/S15
Learning/Community/Live	Skills, rankings, opt-in stream	DONE S16-S18
Quality loop E1+E2	Purchases + lock chef standards	DESIGN

CTO Architecture Snapshot

- Edge: API Gateway routes /api/v1 to 13 domain services (ports 18001-18012)
- Data: PostgreSQL schema-per-domain + PostGIS; Redis Streams + transactional outbox
- Writes never cross schemas; Growth/Ratings read orders cross-schema for evidence only
- PWAs: portal, customer.kitchcu.in, kitchen.kitchcu.in, admin.kitchcu.in
- Full diagrams (architecture, flows, ER): CKAC-COMPLETE-GUIDE.pdf Part III

```
PWAs -> Gateway -> Identity|Catalog|Order|Billing|Marketing|Ratings|
            Growth|Delivery|Learning|Community|Streaming|Notify
         -> PostgreSQL schemas + Redis Streams + MinIO
```

Kitchcu

Kitchcu cloud kitchen platform

hello@kitchcu.in | kitchcu.in | Pilot waitlist open

Full guide: CKAC-COMPLETE-GUIDE.md / .pdf (v2.0) | *CPO blueprint v4.0* | *E1-E2 quality-loop design pack*